

Blackboard Relay: Cooperative Small Language Models via Compressed Latent-State Hand-off

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Abstract

Scaling a single dense model is the dominant route to stronger language capability, but it is expensive and monolithic. We study an alternative for the small-model regime: a team of specialized sub-billion experts that cooperate not through text but by handing off a compressed hidden state—a fixed-size “blackboard” tensor $\mathbf{B} \in \mathbb{R}^{M \times d}$. Building on a 0.5B board-native base model (Cancri-500M, $M=32$, $d=1536$) whose recurrent-memory board is baked in from pre-training, we investigate whether the board can carry information across experts and whether experts forked from a shared base retain a common latent protocol. We report three findings. (i) The board is a high-fidelity channel: it recovers 85% of the full-context perplexity advantage while transmitting only $M=32$ slots. (ii) Treating the board as an augmented state (context compressed into the board, the query kept as exact text) rather than the sole information path fixes a precision-loss failure mode and yields answers that provably originate from the board (control accuracy: board 0.61 vs. empty-board 0.09). (iii) Forking the base into a Understand expert (writes the board) and an Answer expert (reads it) and co-training them through the board attains 0.86 exact-answer accuracy on a multi-clue synthesis task—higher than a single weight playing both roles (0.61) and higher than feeding the full context as text (0.62)—while the empty-board control stays at 0.17. Cross-weight latent alignment thus holds, and specialization is a net win. We release all code and checkpoints, and we are explicit about the current toy-task boundary.

1 Introduction

The prevailing recipe for stronger language models is to grow a single dense network. This is effective but costly, hard to update, and opaque. A complementary question, especially acute for teams operating in the sub-billion-parameter regime, is whether a group of small specialists can be orchestrated to exceed a single model of the same activated scale.

Mixture-of-Experts (MoE) answers a related but different question: it selects token-level experts inside one model with an internal router. We instead consider task-level cooperation between externally separate models, in the spirit of a blackboard system: each expert reads a shared workspace, contributes, and passes control on. The central design choice is the medium of communication. Natural-language hand-off is interpretable but lossy and compute-heavy—every expert must re-encode text it just decoded. We study the alternative advocated by our internal design: experts communicate through a compressed hidden state, a small board tensor $\mathbf{B} \in \mathbb{R}^{M \times d}$, passed serially like a relay baton (Figure 1).

This paper is a focused empirical study of the core mechanism—latent relay—rather than a full multi-expert system. Concretely we ask:

1. Channel. Can a 32-slot board carry cross-segment information at all, on a model never trained for cross-expert relay?
2. Discipline. If the board is used, does the model generate correct answers from it, or merely lower reference perplexity?
3. Alignment. If we fork two different weight sets from a shared base, does the latent board protocol survive the split, and does specialization help?

We answer these in three escalating experiments and report both the positive results and an honest account of where the mechanism currently fails.

2 Related Work

Recurrent memory / state compression. Recurrent Memory Transformers and related work carry information across long contexts through a fixed-size state recycled between segments. Our board is trained in this spirit but is repurposed as an inter-model bus rather than an intra-model memory.

Mixture-of-Experts. MoE routes tokens to experts within a single model and shares one latent space by construction. Blackboard Relay keeps experts as independent checkpoints with an external, task-level schedule; experts can be trained, swapped, and inspected separately.

Latent / soft communication. Soft prompts and prefix tuning show that continuous vectors can steer a frozen model. A recurring finding—which we reproduce—is that a retrofitted soft channel tends to be ignored unless the architecture forces dependence on it. Our base is board-native (the board is present from pre-training), and we additionally use structural bypass blocking as a diagnostic to guarantee the channel is exercised.

Multi-agent LLMs. Text-passing agent pipelines are effective but pay repeated encode/decode costs and accumulate paraphrase drift. We replace the text bus with a latent one and quantify the trade-off.

3 Method

3.1 Board-native base model

Cancri-500M is a 24-layer, $d=1536$ decoder with grouped-query attention (32 query / 8 KV heads), a 32k BPE vocabulary, and a board of $M=32$ slots. The board is not a retrofit: during pre-training each document is processed in segments of length $S=512$ with the board recurred between segments. The elementary operation is a single segment forward,

$$(\mathbf{y}, \mathbf{B}_{\text{out}}) = f_{\theta}([\mathbf{B}_{\text{in}}; \mathbf{x}_{1:S}; \mathbf{q}_w]), \quad (1)$$

where $\mathbf{B}_{\text{in}} \in \mathbb{R}^{M \times d}$ is the incoming board, $\mathbf{x}_{1:S}$ the segment tokens, $\mathbf{q}_w \in \mathbb{R}^{M \times d}$ a set of learned write queries, $\mathbf{y}$ the per-token logits, and $\mathbf{B}_{\text{out}} \in \mathbb{R}^{M \times d}$ the updated board read out from the write positions. We write $\mathbf{B}_{\text{out}}$ only when needed (with_write); $\mathbf{B}_0$ denotes the learned initial board.

3.2 Latent relay

A relay is a composition of Eq. (1) in which only the board crosses the boundary between roles. Given a context c , a query q and an answer a , an Understand pass writes the board and an Answer pass reads it:

$$\mathbf{B}_c = \text{Write}(c, \mathbf{B}_0), \quad (\text{read context} \rightarrow \text{board}) \quad (2)$$

$$\mathbf{y}_a = \text{Read}(\underbrace{[\text{usr } q \text{ ast } a]}_{\text{query as text}}, \mathbf{B}_c). \quad (\text{answer conditioned on board}) \quad (3)$$

The training loss is the cross-entropy on the answer tokens only,

$$\mathcal{L} = -\frac{1}{|\mathcal{A}|} \sum_{t \in \mathcal{A}} \log p_{\theta}(a_t | a_{<t}, q, \mathbf{B}_c), \quad (4)$$

with $\mathcal{A}$ the answer positions (the prompt is masked).

Board as sole path vs. augmented state. Two regimes differ in what the Answer pass sees. Physical bypass blocking removes the query text and forces the board to be the only path—a strong diagnostic that the board is truly read, but lossy for precise tokens. Board-as-augmented-state keeps the query as exact text and lets the board carry the context; text handles precision, the board handles the digested cross-expert state. We use the former to diagnose and the latter for the main results.

Board dropout. With probability ρ we replace $\mathbf{B}_c$ by $\mathbf{B}_0$ during training, so the Answer expert learns to abstain rather than hallucinate when no context is provided. We use $\rho=0.15$.

Algorithm 1: Dual-weight latent-relay training step

Input: context c , query q , answer a ; experts θ_U, θ_A (forked from base); dropout ρ
 $\mathbf{B}_c \leftarrow \text{Write}_{\theta_U}(c, \mathbf{B}_0^U)$; // U writes the board
with prob. ρ : $\mathbf{B}_c \leftarrow \mathbf{B}_0^A$; // board dropout
 $\mathbf{y}_a \leftarrow \text{Read}_{\theta_A}([\text{usr } q \text{ ast } a], \mathbf{B}_c)$; // A reads board + query text
 $\mathcal{L} \leftarrow \text{CE}(\mathbf{y}_a, a)$ on answer positions only;
update θ_U, θ_A with $\nabla_{\theta_U, \theta_A} \mathcal{L}$; // grad flows to both via $\mathbf{B}_c$

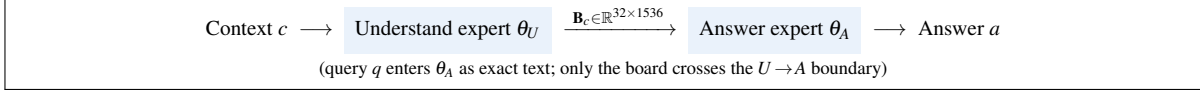


Figure 1: Latent relay between two experts forked from a shared base. Information from the context reaches the answer expert only through the compressed board $\mathbf{B}_c$; the query is provided to the answer expert as exact text.

3.3 Dual-weight experts

The true multi-expert setting uses two weight sets θ_U, θ_A , both forked from the shared base (Figure 1). U only writes the board from context; A only reads it (plus query text) to answer. They are trained jointly by back-propagating Eq. (4) through $\mathbf{B}_c$ into both networks, which forces them to negotiate a shared latent protocol. Algorithm 1 summarizes one step.

4 Experimental Setup

Task. We use a controllable multi-clue synthesis benchmark: (i) attribute lookup—four people each like a distinct color, ask “who likes X ?”; and (ii) two-hop join—a person owns a pet, the pet lives in a city, ask “where does P ’s pet live?”—with distractor facts. The task is deliberately arithmetic-light (the 0.5B base is weak at arithmetic) and requires reading the context: the answer is not inferable from the query alone. Contexts are short, answers are one sentence with a single decisive key token (a name or city).

Metrics. Our primary metric is exact-answer accuracy: greedy decoding from the board, scored as correct iff the gold key token appears. Because teacher-forced perplexity is diluted by echoed tokens, we treat perplexity-based board lift $(\text{ppl}_{\text{null}} - \text{ppl}_{\text{board}})$ and recovery $= (\text{ppl}_{\text{null}} - \text{ppl}_{\text{board}}) / (\text{ppl}_{\text{null}} - \text{ppl}_{\text{full}})$ as secondary. We always report three controls on a held-out generation seed: board (relayed board + query), null (empty board + query, the prior/echo baseline), and full (full context as text + query, the upper bound). A large board–null gap certifies that answers originate from the board.

Training. AdamW ($\beta=0.9, 0.95$), lr 10^{-5} with 3% warmup and cosine decay, gradient clip 1.0, bfloat16, effective batch 16, 3 epochs over 2×10^4 synthetic examples ($\sim 3.7\text{k}$ steps, $\approx 10\text{--}15$ min on one MI300X). Dual-weight training holds two 0.5B models resident simultaneously.

5 Results

The board is a high-fidelity channel. Even zero-shot—with no relay training—compressing the context into 32 board slots recovers 85% of the perplexity advantage of reading the full context as text (ppl: 13.0 $\rightarrow$ 4.25 vs. full-text 2.70; Table 1). The physics of latent compression works.

Correct generation requires relay training; and precision is lossy. A high board-lift does not imply correct generation. Under sole-path (bypass-blocked) decoding the base model reproduces the topic and format of an answer but corrupts precise operands: on a word problem it recovers the operands yet flips an operation or adds a spurious step; on a two-attribute problem it drops the exact numbers and degrades to an empty template. Moving to board-as-augmented-state (context in the board, query as exact text) and finetuning for the relay objective (Eq. 4) raises exact-answer accuracy

Table 1: Held-out exact-answer accuracy (generation seed unseen in training) and the perplexity channel diagnostic. board—null is the certified board contribution. Dual-weight experts beat both the single shared weight and the full-text upper bound.

Configuration	board $\uparrow$	null	full	board—null $\uparrow$
Cancr-500M (zero-shot)	0.150	0.000	0.455	+0.150
Single weight (relay-finetuned)	0.610	0.085	0.575	+0.525
Dual weight ($U+A$)	0.859	0.172	0.625	+0.687

Perplexity channel (zero-shot, board-as-augmented-state):
 $\text{ppl}_{\text{null}}=13.0$, $\text{ppl}_{\text{board}}=4.25$, $\text{ppl}_{\text{full}}=2.70 \Rightarrow \text{recovery} = 0.85$

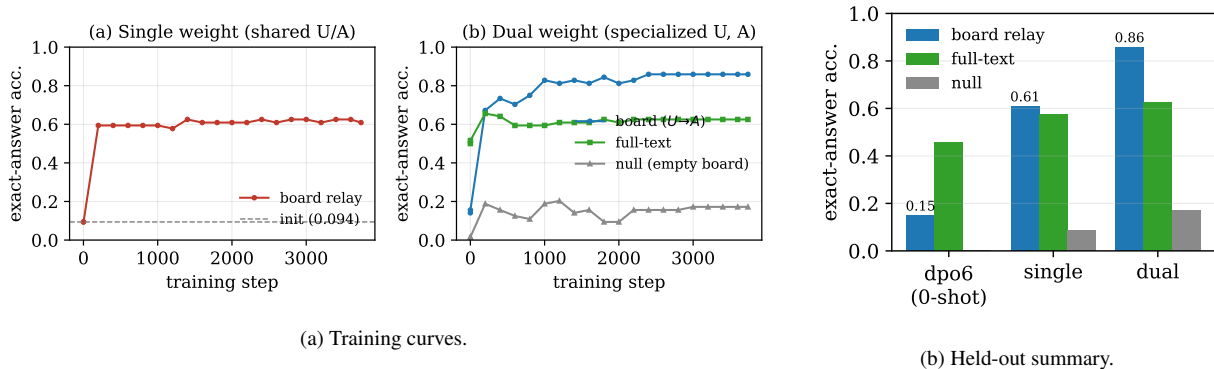


Figure 2: (a) Exact-answer accuracy during training. A single weight playing both roles plateaus near 0.62; two specialized weights reach 0.86 while the empty-board control (null) stays low. (b) Board relay vs. full-text vs. null across the three configurations on the held-out seed.

from 0.09 to 0.61 (Figure 2aa). The null control (0.085) confirms the answer is decoded from the board, not guessed, and the result generalizes to a held-out generation seed.

Cross-weight alignment holds; specialization wins. Forking the base into separate U and A weights and co-training through the board raises accuracy to **0.859** (Figure 2ab, Table 1), above both the single shared weight (0.61) and the full-text upper bound (0.625), while null stays at 0.17 (gap +0.687). Two conclusions follow. First, the latent protocol survives the split: two independent weight sets, connected only by a 32-slot tensor, learn to write and read a common board—direct support for the requirement that experts fork from a shared base. Second, specialization is a net win: a weight dedicated to one role outperforms a weight juggling both, and for a small model a pre-digested board beats in-context retrieval over distractors.

6 Discussion

Why does the board beat full text for a 0.5B model? Full-text answering forces the answer network to perform retrieval over distractor sentences in-context, which a small model does poorly. A dedicated Understand expert instead pre-digests the context into a clean fixed-size summary that the Answer expert was explicitly trained to read. This is precisely the Blackboard-Relay thesis: a pipeline of specialists with a dense, learned interface can beat a single generalist doing everything in one pass—at least where the interface is well matched to the sub-task. The same view explains the sole-path failure: a 32-slot board is a lossy code for exact operands, so tasks that need verbatim precision should keep a text path (as our augmented-state regime does) rather than force everything through the latent bus.

7 Limitations

We are deliberate about scope. (i) The benchmark is a controllable synthetic task over a small closed vocabulary; the architectural conclusions (cross-weight alignment, specialization gain) are supported, but the magnitude of “board

beats full-text” will depend on task complexity, context length, and base capability and should not be extrapolated to open-domain tasks. (ii) Precise content remains the board’s weak spot; we sidestep rather than solve it. (iii) Perplexity-based board-lift is diluted by echoed tokens, so we rely on generation accuracy; the residual null accuracy (0.17) indicates minor prior leakage. (iv) We have not yet evaluated diverse real data, chains longer than two experts, or heterogeneous expert capacities. These are the immediate next steps.

8 Conclusion

We showed, on a board-native 0.5B model, that small language models can cooperate by handing off a compressed hidden state rather than text. The 32-slot board is a high-fidelity channel (0.85 recovery); treating it as an augmented state fixes a precision-loss failure and yields answers provably decoded from the board; and forking a shared base into an Understand and an Answer expert preserves the latent protocol and improves accuracy to 0.86, beating both a single dual-role weight and the full-text baseline. Latent relay between forked small experts is therefore viable, and specialization pays. Future work targets real, diverse data, longer expert chains, and closing the precision gap via verifiable-selection decoding.

Outlook. We view the two-expert relay as the seed of a broader architecture. If a compressed board can carry state between two forked experts, it can in principle serve a swarm of many small specialists that read from and write to a shared latent workspace—a global blackboard—rather than a single monolith. Such a design echoes the global-workspace view of cognition and, loosely, the way populations of neurons in the brain coordinate through a shared medium rather than point-to-point text: capability would emerge from the composition and scheduling of many cheap, swappable experts over a common latent bus. Whether this scales beyond toy tasks—and how to route, schedule, and verify contributions on the board—is the direction this work is meant to open.

Reproducibility

All scripts and checkpoints are released: the relay diagnostic (`cancri_relay.py`), the augmented-state harness and finetuner (`cancri_relay2.py`, `cancri_relay2_ft.py`), and the dual-weight trainer with a gradient-connectivity check (`cancri_brp_dual.py`). Checkpoints: `expert_understand.ckpt`, `expert_answer.ckpt`. The multi-clue benchmark is generated procedurally with a fixed seed.

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